

King Saud University

College of Computer and Information Sciences

Department of Computer Science

CSC 227: Operating Systems Programming Project 2: Contiguous memory allocation

Simulator

First Semester 1447

Student Name	Student ID
لين الزهراني	444200939
رذاذ بن راكان	444200451
رزان المطيري	444200495
منيره الحويكان	444200541
سارة الصبيحي	444200926

Task Distribution Details

Name	Task
لين الزهراني	Memory Initialization
رذاذ بن راكان	Allocation Algorithms
رزان المطيري	Compaction & Display
منيره الحويكان	Release Function
سارة الصبيحي	Testing & Documentation

Scenario 1 — First Fit Allocation

Memory size: 500

Input:

- Process A, size 120, strategy F
- Process B, size 50, strategy F
- Process C, size 200, strategy F

Explanation:

First Fit places each process in the first available free block fitting the size.

```
Enter the total memory size:
500

Memory initialized successfully!

Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 1
Enter process name: A
Enter size: 120
Enter strategy (F/B/W): F
Process A allocated from 0 to 119

Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 1
Enter process name: B
Enter size: 50
Enter strategy (F/B/W): F
Process B allocated from 120 to 169

Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 1
Enter process name: C
Enter size: 200
Enter strategy (F/B/W): F

Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 4

Memory status:
-----
Addresses [0 : 119]      Process A
Addresses [120 : 169]   Process B
Addresses [170 : 369]   Process C
Addresses [370 : 499]   Unused (Free)
-----
```

Scenario 2 — Release Function

Input:

- release B

Explanation:

B is removed and the block becomes free.

```
-----  
  
Memory Allocation Simulator  
---  
1. Request (Allocate) memory  
2. Release memory  
3. Compact memory  
4. Display memory status  
5. Exit  
---  
Enter your choice: 2  
Enter process name to release: B  
Process B released.
```

Scenario 3 — Best Fit

Input:

- Process D, size 30, strategy B

Explanation:

Best Fit chooses the *smallest* free block that fits size 30.

```
Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 1
Enter process name: D
Enter size: 30
Enter strategy (F/B/W): B
Process D allocated from 120 to 149
```

Scenario 4 — Worst Fit

Input:

- Process E, size 80, strategy W

Explanation:

Worst Fit selects the *largest* free block.

```
Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 1
Enter process name: E
Enter size: 80
Enter strategy (F/B/W): W
Process E allocated from 370 to 449
```

Scenario 5 — Compact

Input:

- Compact memory

Explanation:

All allocated blocks are shifted to the beginning; all free space becomes one block at the end.

```
Memory Allocation Simulator
---
1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit
---
Enter your choice: 3
Memory compacted successfully.
```

Scenario 6 — Final Display

Memory Allocation Simulator

1. Request (Allocate) memory
2. Release memory
3. Compact memory
4. Display memory status
5. Exit

Enter your choice: 4

Memory status:

Addresses [0 : 119] Process A

Addresses [120 : 149] Process D

Addresses [150 : 349] Process C

Addresses [350 : 429] Process E

Addresses [430 : 499] Unused (Free)
